Functional Requirements Document

(FRD)

Small Business Order & Inventory Management System

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1. Introduction

Purpose

This Functional Requirements Document (FRD) defines how the **Small Business Order & Inventory Management System** should function to satisfy the business requirements identified in the BRD.

The document describes the main system functions, user interactions, business rules, validations, and expected system behavior.

2. System Users

The system will support the following users:

Table 1: Users and main functions

User	Main Functions
Business Owner	View dashboard, sales and inventory information
Admin	Manage products, inventory, customers and orders
Sales Staff	Create and manage customer orders
Inventory Staff	View and update inventory

3. Functional Requirements

User Authentication & Access

FR-01: User Login

The system shall allow registered users to log in using valid credentials.

Input:

- Username / Email
- Password

Expected Result:

- Valid credentials → User is logged in.
- Invalid credentials → System displays an error message.

FR-02: Role-Based Access

The system shall display functions according to the user's assigned role.

For example, sales staff should not be able to access administrative functions unless permission is provided.

4. Product Management

FR-03: Add Product

The system shall allow authorized users to add a new product.

Required information:

- Product name
- Category
- Unit price
- Initial stock quantity
- Minimum stock level

FR-04: Edit Product

The system shall allow authorized users to update product information.

FR-05: Search Product

The system shall allow users to search for products using the product name or category.

FR-06: Deactivate Product

The system shall allow authorized users to deactivate products that are no longer available.

5. Inventory Management

FR-07: View Inventory

The system shall display:

- Product name
- Current stock quantity
- Minimum stock level
- Stock status

FR-08: Update Stock

Authorized inventory users shall be able to update stock quantities.

The system shall record the updated stock quantity.

FR-09: Check Stock Availability

When creating an order, the system shall automatically check whether sufficient stock is available.

FR-10: Low-Stock Alert

The system shall display a low-stock alert when:

Current Stock ≤ Minimum Stock Level

6. Customer Management

FR-11: Add Customer

Authorized users shall be able to create a customer record containing:

- Customer name
- Contact number
- Email (optional)
- Address (optional)

FR-12: Edit Customer

Authorized users shall be able to update customer information.

FR-13: View Customer Order History

The system shall allow authorized users to view previous orders associated with a customer.

7. Order Management

FR-14: Create Order

Sales staff shall be able to create a new customer order.

The order shall include:

- Customer

- Product
- Quantity
- Unit price
- Total amount
- Order date

FR-15: Validate Stock Before Order Confirmation

Before confirming an order, the system shall compare the requested quantity with the available stock.

If sufficient stock exists:

Order can be confirmed.

If insufficient stock exists:

Order cannot be confirmed.

System displays an appropriate message.

FR-16: Update Order Status

Authorized users shall be able to update the order status.

Available statuses:

- Pending
- Processing
- Ready
- Completed
- Cancelled

FR-17: View Orders

Users with permission shall be able to view orders and filter them by relevant information such as:

- Order ID
- Customer
- Order status
- Date

FR-18: Cancel Order

Authorized users shall be able to cancel an order according to the business rules.

If stock was deducted for the order, the system shall restore the relevant stock quantity when the cancellation is confirmed.

8. Dashboard & Reports

FR-19: View Dashboard

The system shall provide a dashboard containing basic business information such as:

- Total orders
- Pending orders
- Completed orders
- Total sales
- Low-stock products

FR-20: Sales Information

The system shall allow authorized users to view basic sales information based on a selected period.

FR-21: Inventory Information

The system shall allow authorized users to view current inventory information and identify low-stock products.

9. Business Rules

Table 2: Business rules

ID	Rule
RULE-01	An order cannot be confirmed when requested quantity is greater than available stock.
RULE-02	Stock quantity cannot be negative.
RULE-03	Low-stock status applies when current stock is less than or equal to the minimum stock level.

RULE-04	Only authorized users can modify product and inventory information.
RULE-05	A completed order cannot be edited unless authorized by the business rules.
RULE-06	Cancelling an order after stock deduction should restore the deducted quantity.

10. Validation Requirements

The system shall perform the following basic validations:

- Required fields cannot be empty.
- Product quantity must be greater than zero.
- Product price cannot be negative.
- Stock quantity cannot be negative.
- Customer contact information should follow the accepted format.
- Duplicate product records should be prevented where applicable.
- An order cannot be confirmed without a valid customer and product.
- An order cannot be confirmed when sufficient stock is unavailable.

11. Main Use Cases

Table 3: Use cases

Use Case ID	**Use Case**	**Primary User**
UC-01	Login	All Users
UC-02	Manage Products	Admin
UC-03	Manage Inventory	Admin / Inventory Staff
UC-04	Manage Customers	Admin / Sales Staff
UC-05	Create Order	Sales Staff
UC-06	Manage Orders	Admin / Sales Staff
UC-07	View Low-Stock Alerts	Admin / Inventory Staff
UC-08	View Dashboard	Business Owner / Admin

UC-09	View Reports	Business Owner / Admin

12. Key Use Case Description

UC-05: Create Customer Order

Primary Actor: Sales Staff

Precondition:
The staff member is logged into the system.

Main Flow:

1. Staff selects "Create Order".
2. Staff selects or searches for the customer.
3. Staff selects the required product.
4. Staff enters the required quantity.
5. System checks product availability.
6. System calculates the order total.
7. Staff confirms the order.
8. System creates the order.
9. System updates the inventory.
10. System displays the order confirmation.

Alternative Flow:

If sufficient stock is unavailable:

1. System identifies insufficient stock.
2. System displays an insufficient-stock message.
3. Order confirmation is prevented.
4. Staff can modify the quantity or select another product.

Postcondition:

- Order is recorded.
- Inventory is updated.

- Order receives an initial status.

13. Non-Functional Requirements

NFR-01 – Usability

The system should provide a simple and user-friendly interface.

NFR-02 – Performance

Common operations such as login, product search, and order creation should respond within a reasonable time.

NFR-03 – Security

Users should only access functions permitted by their roles.

NFR-04 – Reliability

The system should maintain accurate order and inventory information during normal operation.

14. Requirements Traceability

The functional requirements should be traceable to the business requirements defined in the BRD.

Table 4: Business requirements and functional requrements

Business Requirement	**Functional Requirements**
BR-01 Product Management	FR-03, FR-04, FR-05, FR-06
BR-02 Inventory Management	FR-07, FR-08, FR-09, FR-10
BR-03 Order Management	FR-14, FR-15, FR-16, FR-17, FR-18
BR-04 Stock Availability	FR-09, FR-15
BR-05 Order Status	FR-16, FR-17
BR-06 Customer Management	FR-11, FR-12, FR-13
BR-07 Low Stock Alerts	FR-10
BR-08 Sales & Inventory Information	FR-19, FR-20, FR-21
BR-09 Role-Based Access	FR-02